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Department of Mathematics

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Math 208 Major Exam I Term 243

Original exam date: 3 July 2025. Practice version: questions and choices shuffled.

Net time allowed: 90 minutes

1. If the acceleration of a particle is $a(t) = 18 \cos(3t)$, the initial velocity is $v_0 = 4$, and the initial position is $x_0 = -7$, then $x\left(\frac{\pi}{2}\right) =$

- (a) $2\pi - 7$
- (b) $2\pi - 5$
- (c) $2\pi + 7$
- (d) $2\pi + 5$
- (e) $7 - 2\pi$

2. By making a suitable substitution, the differential equation $x^2 y' + 2xy = 5y^4$ can be transformed into the linear differential equation

- (a) $v' + \frac{6}{x}v = -\frac{14}{x^2}$
- (b) $v' - \frac{3}{x}v = -\frac{12}{x^2}$
- (c) $v' - \frac{6}{x}v = -\frac{15}{x^2}$
- (d) $v' - \frac{6}{x}v = -\frac{20}{x^2}$
- (e) $v' + \frac{6}{x}v = -\frac{10}{x^2}$

3. By making a suitable substitution, the differential equation $xy' = y + \sqrt{x^2 + y^2}$ can be transformed into the separable differential equation

- (a) $\sqrt{1 + v^2} dv = x dx$
- (b) $\frac{1}{\sqrt{1 + v^2}} dv = x^2 dx$
- (c) $\frac{1}{\sqrt{1 + v^2}} dv = \frac{1}{x} dx$
- (d) $\frac{1}{\sqrt{1 + v^2}} dv = \frac{1}{x^2} dx$
- (e) $\sqrt{1 + v^2} dv = \frac{1}{x} dx$

4. If the differential equation $(5x^4 - 4x^3y^3) dx + (5 + kx^4y^2) dy = 0$ is an exact differential equation, then $k =$

- (a) -6
- (b) -7
- (c) -3
- (d) -4
- (e) -5

5. Which one of the following subsets is a subspace of $\mathbb{R}^4$?

- (a) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 + x_2 = 1$
- (b) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 = 1$
- (c) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1x_2 = 0$
- (d) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1 = 0$
- (e) The set of all vectors (x_1, x_2, x_3, x_4) such that $x_1x_3 = 0$

6. A culture initially has P_0 bacteria. At $t = 1$ hour the number of bacteria is measured to be $\frac{3}{2}P_0$. If the rate of growth is proportional to the number of bacteria $P(t)$ present at time t , the time necessary for the number of bacteria to triple is

- (a) $\frac{\ln 2}{\ln 5 - \ln 4}$
- (b) $\frac{\ln 3}{\ln 3 - \ln 2}$
- (c) $\frac{\ln 3}{\ln 4 - \ln 3}$
- (d) $\frac{\ln 3}{\ln 3 + \ln 2}$
- (e) $\frac{\ln 3}{\ln 7 - \ln 5}$

7. The sum of all values of r such that $y = x^r$ is a solution of the differential equation $3x^2y'' + 6xy' + y = 0$ is

- (a) -1
- (b) -3
- (c) -5
- (d) -4
- (e) -2

8. The general solution of the linear differential equation $xy' = 2y + x^3 \cos x$ is given by

- (a) $y(x) = x^2(\cos x + c)$
- (b) $y(x) = x^3(\sin x + c)$
- (c) $y(x) = x^3(\cos x + c)$
- (d) $y(x) = x(\sin x + c)$
- (e) $y(x) = x^2(\sin x + c)$

9. Let $\mathbf{u} = \langle 2, 3 \rangle$, $\mathbf{v} = \langle -4, 5 \rangle$, $\mathbf{w} = \langle -16, 42 \rangle$ be vectors in $\mathbb{R}^2$. If $\mathbf{w} = a\mathbf{u} + b\mathbf{v}$, then $a + b =$

- (a) 10
- (b) 4
- (c) 6
- (d) 8
- (e) 9

10. A general solution of the differential equation $y' = (9x + 4y)^2$ is given by

- (a) $y = \frac{3}{5} [\tan(6x + c) + 6x]$
- (b) $y = \frac{3}{8} [\tan(6x + c) - 6x]$
- (c) $y = \frac{1}{8} [\tan(6x + c) + 6x]$
- (d) $y = \frac{1}{8} [\tan(6x + c) - 6x]$
- (e) $y = \frac{1}{5} [\tan(6x + c) - 6x]$

11. A general solution of the exact differential equation $(e^{2y} - y \cos(xy)) dx + (2xe^{2y} - x \cos(xy) + 2y) dy = 0$ is

- (a) $2xe^{2y} - \sin(xy) + y^2 = c$
- (b) $xe^{2y} + y \sin(xy) + y^2 = c$
- (c) $xe^{2y} - x \sin(xy) + y^2 = c$
- (d) $xe^{2y} - \sin(xy) + y^2 = c$
- (e) $3xe^{2y} + y \sin(xy) + y^2 = c$

12. The solution of the initial-value problem $(x^2 - 9)\frac{dy}{dx} = 12y$, $y(0) = 4$ is

(a) $y = 4 \left(\frac{x-3}{x+3} \right)^2$

(b) $y = -4 \left(\frac{x-2}{x+2} \right)^3$

(c) $y = -4 \left(\frac{x-3}{x+3} \right)^3$

(d) $y = 4 \left(\frac{x-3}{x+3} \right)^4$

(e) $y = 4 \left(\frac{x-2}{x+2} \right)^2$

13. The solution space of the system

$$x_1 - 2x_2 + 2x_3 + x_4 = 0$$

$$x_1 - x_2 + 2x_3 + 5x_4 = 0$$

$$x_1 - 4x_2 + 2x_3 - 7x_4 = 0$$

is the set of all linear combinations of the form $s\mathbf{u} + t\mathbf{v}$, where s and t are real numbers and

(a) $\mathbf{u} = (-2, 0, 1, 0)$ and $\mathbf{v} = (-9, -4, 0, 1)$

(b) $\mathbf{u} = (-2, 0, -1, 0)$ and $\mathbf{v} = (9, -4, 0, 1)$

(c) $\mathbf{u} = (-2, 0, 1, 0)$ and $\mathbf{v} = (-9, -4, 0, -1)$

(d) $\mathbf{u} = (2, 0, 1, 0)$ and $\mathbf{v} = (9, -4, 0, 1)$

(e) $\mathbf{u} = (-2, 0, 1, 0)$ and $\mathbf{v} = (9, 4, 0, 1)$

14. Let $\mathbf{v} = (2, 5, -4)$ and $\mathbf{w} = (1, -2, -3)$ be two vectors in $\mathbb{R}^3$. Find $|\mathbf{v} - 2\mathbf{w}|$.

(a) $\sqrt{17}$

(b) $\sqrt{87}$

(c) 9

(d) $\sqrt{83}$

(e) $\sqrt{85}$

15. The value of k for which the vectors $\mathbf{u} = (1, 0, 2)$, $\mathbf{v} = (-1, 2, k)$ and $\mathbf{w} = (0, 9, 3)$ of $\mathbb{R}^3$ are linearly dependent is

(a) $\frac{3}{4}$

(b) $\frac{4}{5}$

(c) $-\frac{4}{3}$

(d) $-\frac{2}{3}$

(e) $\frac{2}{3}$

Answer Key

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Q	Ans	Brief explanation
1	(b)	$v = 6 \sin 3t + 4$, $x = -2 \cos 3t + 4t - 5$; $\cos \frac{3\pi}{2} = 0$ gives $2\pi - 5$.
2	(c)	Bernoulli $n = 4$, $v = y^{-3}$: $-\frac{1}{3}v' + \frac{2}{x}v = \frac{5}{x^2}$.
3	(c)	$v = y/x$: $xv' = \sqrt{1 + v^2}$.
4	(c)	$M_y = -12x^3y^2 = N_x = 4kx^3y^2$, so $k = -3$.
5	(d)	$x_1 = 0$ is homogeneous linear; the others miss 0 or fail addition.
6	(b)	$k = \ln \frac{3}{2}$; $e^{kt} = 3$ gives $t = \frac{\ln 3}{\ln 3 - \ln 2}$.
7	(a)	$3r(r-1) + 6r + 1 = 3r^2 + 3r + 1 = 0$; roots sum to -1 .
8	(e)	$\rho = x^{-2}$: $(x^{-2}y)' = \cos x$, so $y = x^2(\sin x + c)$.
9	(a)	$2a - 4b = -16$, $3a + 5b = 42$ give $a = 4$, $b = 6$.
10	(b)	$v = 9x + 4y$: $v' = 9 + 4v^2$, $v = \frac{3}{2} \tan(6x + c)$, $y = \frac{v-9x}{4}$.
11	(d)	$F = xe^{2y} - \sin(xy) + g(y)$ with $g' = 2y$.
12	(a)	Partial fractions: $\ln y = 2 \ln \left \frac{x-3}{x+3} \right + C$; $y(0) = 4$ gives $K = 4$.
13	(a)	$x_2 = -4t$, $x_1 = -2s - 9t$ with $x_3 = s$, $x_4 = t$.
14	(e)	$\mathbf{v} - 2\mathbf{w} = (0, 9, 2)$, length $\sqrt{85}$.
15	(c)	$\det = -12 - 9k = 0$, so $k = -\frac{4}{3}$.